

VLSI Industry Offers Mostly Verification Based Jobs

Today, VLSI devices are found everywhere around us. We find advanced VLSI chips in our cars, cell phones, household appliances, cameras, medical devices and many other places. This rapidly evolving sector offers exciting opportunities for those with strong fundamentals in electronic circuit design and hardware description languages, interest in VLSI design and verification and, more importantly, the skill to put know-how of VLSI concepts to practice



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With the advances being made in technologies like process geometries, feature and product innovations on a daily basis, there is a constant need to design, develop and re-engineer integrated circuits (ICs). Since products like mobile phones are being released with new features in increasingly shorter cycles, there is a healthy demand for qualified very large scale integration (VLSI) engineers to work on these products. Therefore there is good scope for a career in the VLSI industry.

The VLSI field offers exciting growth opportunities for engineers who are strong in electronic design fundamentals, have an interest in VLSI design and verification and

know how to apply VLSI concepts to practice. Harish Mysore, director, India operations, Global IEEE Institute for Engineers Private Limited, informs, "In the VLSI chip design and verification alone, 20,000 to 30,000 engineers are currently employed with over 200,000 engineers working in the broader semiconductor industry, including embedded systems development and board-level hardware design."

According to Vivek Madhukar, COO, TimesJobs.com, VLSI professionals are always in high demand in the fast-changing chip designing industry. There are over 150 companies catering to this industry, including big names like Texas Instruments, Infineon, Freescale Semiconductor, Cadence, HCL, Intel, Lucent, Motorola, Philips

Semiconductor, Qualcomm, Sasken, Conexant, Wipro and TCS, to name a few. He says, "A career in one of these companies is highly sought after and developing VLSI skills is a good way to make this dream a reality."

It is not easy to get into design

On the flip side, scope for a career in the design side of VLSI industry is currently somewhat limited as compared to other areas, feels Subhajit Sen, associate professor, International Institute of Information Technology, Bangalore, for three particular reasons. One, VLSI or chip design requires a deeper level of knowledge and skills than other electronics related fields. Two, VLSI chip design is expensive and requires access to high-cost, specialised electronic design

The expectations in terms of the know-how of analogue complementary metal oxide semiconductor (CMOS) designs/issues are very high, which makes it difficult to get into analogue based VLSI companies. Most companies prefer to take only M.Techs from reputed institutes such as NITs/IITs